🏛️ CampusFix

Campus Maintenance & Infrastructure Triage Platform — Official Hackathon Documentation

Production Live Prototype:
Launch Live Application

Developer Sandbox URL:
View Developer Build

📌 Project Overview & Core Mission

The Problem

Traditional university and corporate campus maintenance tracking is fundamentally fragmented and slow. Critical faults like failed Wi-Fi access points, fluctuating voltage regulators, structural issues, and utility leaks are typically handled through manual, non-standardized communication pipelines (untracked emails, paper forms, or verbal reporting). This leads to unmonitored SLA breaches, major administrative backlogs, and duplicated reports due to zero public visibility.

The Solution

CampusFix introduces an automated, high-performance maintenance orchestration platform. Powered by server-side **Gemini AI** intelligence, it immediately triages, tags, scores for urgency, and dispatches incident tickets to isolated specialist lines. By dividing the workspace cleanly between an interactive **Reporter Panel** and an analytical **Authority Command Dashboard**, operational clarity is instantly restored.

🔑 Core Feature Architecture

1. 📢 Frictionless Reporter Panel

- **Granular Structural Metadata:** Users can effortlessly input precise details including architectural block, floor level, specific room identifiers, and plain-text problem descriptions.
- **Base64 Cryptographic Evidence Pipeline:** Eliminates storage overhead during fast workflows by capturing and streaming live photographic logs directly into the incident schema.
- **Real-Time Lifecycle Feedback:** Immediate visual reflections show exactly when the item enters the server-side analysis pipeline.

2. 🛡️ Intelligent Authority Command Dashboard

- **Autonomous Gemini AI Triage Engine:** No human interaction required for sorting. The incoming report text is instantly parsed on the backend. The AI determines target operational departments, calculates a 4-tier urgency score (**Low** **Medium** **High** **Critical**), structures actionable technical task descriptions, and adds standard diagnostic telemetry.

- **Contamination-Free Department Queues:** Beautifully isolated task lines cleanly partition work orders into specialized pools: *IT Support, Plumbing, Electrical, Carpentry, and HVAC*.
- **Live SLA Diagnostic Telemetry:** Renders visual resolution progress indicators, dynamic ticket closure ratios, and ongoing health charts.

3. 🎨 Visual Craftsmanship & Premium UI/UX

- **Translucent Glassmorphic Design:** Tailored using advanced Tailwind primitives combined with high-contrast ambient glow rings, micro-interactions, and beautiful structural layouts.
- **Rapid Testing Bypass Modules:** Includes pre-configured developer sandbox profiles and integrated single-click authentication protocols to ensure lightning-fast evaluation by judges.

🛠️ Full-Stack Technology Profile

LAYER	TECHNOLOGIES EMPLOYED & STRATEGIC PURPOSE
Core UI Architecture	React 18 paired with TypeScript for absolute component type safety.
Build Automation	Vite Pipeline ensuring sub-second Hot Module Replacement (HMR) and optimized build bundles.
Styling Core	Tailwind CSS leveraging customized theme modules for unified design tokens.
Motion Engine	Framer Motion powering fluid state transformations and dynamic layout queues.
AI Engine	Official Google Gemini Server SDK (`@google/genai`) optimizing prompt structures.
Secure Middleware	Express.js Proxy API ensuring high-performance server token abstraction.
State & Persistence	Atomic persistent JSON-backed local storage tracking operational states reliably.

⚙️ System Architecture & Data Lifecycle

The flow below describes how user incident inputs move securely through our server proxy down to the autonomous AI evaluation layer:

```
[Reporter App UI] ──(Payload + Base64 Image)──> [Secure Express.js Proxy Server] | (Appends HIDDEN API Key) ▼ [Authority Dashboard] <──(JSON State) <── [JSON DB] <── [Gemini Triage Model]
```

🚧 Technical Challenges & Resolutions

Challenge 1: Securing Private API Keys in Client-Side Architectures

Hurdle: Calling the Google Gemini SDK directly from client-side React components would bundle the project's secret API credentials into open distribution code, creating critical security vulnerabilities.

Resolution: Engineered a robust **Full-Stack Proxy Architecture**. We abstracted the Gemini SDK behind an isolated Express.js backend layer. The secret key is stored strictly as a server-side environment variable (`process.env.GEMINI_API_KEY`). Client modules interact safely using standardized local endpoints (`/api/reports`).

Challenge 2: Horizontal Scrolling & UI Overflow Refinement

Hurdle: Rendering comprehensive multi-department category switches on cramped mobile layouts caused broken grids and awkward vertical stack stretches.

Resolution: Programmed an optimized horizontal touch-swipe element using native Tailwind overflow properties. Built reactive DOM references to ensure seamless navigation across varying screen dimensions.

🔮 Project Roadmap & Future Scope

- 📟 **Automated Dispatch Escalation:** Integrating Twilio SMS workflows to automatically trigger instant cellular dispatches directly to engineers on *Critical* tickets.
- 📍 **Vector CAD/SVG Interactive Floor-Plans:** Interactive geospatial mapping allowing administrators to view live pins placed over the actual physical blueprints of the campus.
- 📈 **Predictive Facilities Lifecycle Analytics:** Mining historical ticket trends to flag infrastructure arrays showing signs of imminent degradation.

👨‍💻 Engineering & Product Design

Developed with absolute precision and dedication to engineering standard excellence by **Sarfarosh Alam**.

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